

Invited review

Myelin plasticity in the central nervous system

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ABSTRACT

Myelin sheaths, specialized segments of oligodendrocyte (OL) plasma membranes in the central nervous system (CNS), facilitate fast, saltatory conduction of action potentials down axons. Changes to the fine structure of myelin in a neural circuit, including sheath thickness and internode length (length of myelin segments between nodes of Ranvier), are expected to affect conduction velocity of action potentials. Myelination of the mammalian CNS occurs in a stereotyped, progressive pattern and continues well into adulthood in humans. Recent evidence from zebrafish, rodents, non-human primates, and humans suggests that myelination may be sensitive to experiences during development and adulthood, and that varying levels of neuronal activity may underlie these experience-dependent changes in myelin and myelin-forming cells. Several cellular, molecular, and epigenetic mechanisms have been investigated as contributors to myelin plasticity. A deeper understanding of myelin plasticity and its underlying mechanisms may provide insights into diseases involving myelin damage or dysregulation.

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1. Introduction

In recent decades, a growing body of evidence has come to light illustrating that myelin and the cells that form it are dynamic and responsive to the neurons they ensheath. Myelin plasticity refers to the response of myelin-forming cells to neuronal activity. In the healthy brain, this process appears to be adaptive. Here, we review the emerging field of activity-regulated myelin plasticity and discuss what is understood about the cellular and molecular mechanisms that underlie adaptive changes in myelin in the developing and adult central nervous system (CNS). We further consider how adaptive myelination may differ between regions within the CNS and how knowledge of this underappreciated aspect of neural plasticity may inform pressing clinical problems in neuroscience.

2. Myelination

Myelination in the CNS refers to the ensheathment of axons by the plasma membranes of oligodendrocytes (OLs) in a spiral fashion (wrapping) with compaction into segments of compact myelin to form internodes. Internodal segments alternate with nodes of Ranvier (unwrapped axonal membrane), where an accumulation of voltage-gated sodium channels allows for action potential propagation (Sherman and Brophy, 2005). Immediately adjacent to nodes of Ranvier on either side are segments of paranodal myelin, where interactions between cell adhesion molecules and the extracellular matrix help to define and may initiate the formation of nodes of Ranvier (Susuki and Rasband, 2008). Beyond the paranodes are juxtaparanodal regions of myelin, which express voltage-gated potassium channels that are thought to maintain neuronal resting potential in internodal segments and to mediate axoglial communication (Poliak and Peles, 2003). Myelin insulates the internodal regions to decrease membrane capacitance and increase membrane resistance, thereby facilitating fast saltatory propagation of action potentials (Hartline and Colman, 2007). The effect myelin has on the speed of conduction depends upon the thickness of the myelin sheath relative to the diameter of the axon as well as to the length of the internodal segment, as discussed in more detail below. Internodal segment length changes over the lifespan (Young et al., 2013) and can vary along a single axon depending on that axon's regional location (Tomassy et al., 2014).

In addition to generating myelin sheaths, OLs appear to provide metabolic support for the axons they ensheath. Products of OL glycolysis (lactate or pyruvate) may diffuse through monocarboxylate transporters in paranodal myelin into underlying axons, fueling mitochondrial metabolism and promoting axonal function integrity (Saab et al., 2013); a similar axonal metabolic support function has been described for Schwann cells, the myelinating cells of the peripheral nervous system (PNS) (Brown et al., 2012). Loss of myelin-specific proteins such as CNP1 (Lappe-Siefke et al., 2003) or conditional knockout of mitochondrial cytochrome c oxidase in OLs (Fünfschilling et al., 2012) has been shown to cause metabolic dysregulation in axons while sparing myelin structure, suggesting that axon-OL metabolic coupling is important for neuronal metabolic function.

2.1. Cellular effectors of adaptive myelination

Oligodendrocytes, the myelin-producing cells of the mammalian CNS, are generated by oligodendrocyte precursor cells (OPCs), a largely unipotent progenitor cell that can divide symmetrically to produce two daughter OPCs or divide asymmetrically to self-renew while producing one daughter OL (Hughes et al., 2013; Zhu et al., 2011). OPCs remain proliferative throughout life, dividing in adult mice every 20–40 days on average when raised in a non-enriched environment (Nishiyama et al., 2009; Kang et al., 2010; Simon et al., 2011; Young et al., 2013).

Despite the inherent capacity of forebrain OPCs for self-renewal (Psachoulia et al., 2009), a population of OPCs derived from SVZ progenitors has been found in the postnatal rodent forebrain (Levison and Goldman, 1993), mobilizing especially after demyelinating injury (Nait-Oumesmar et al., 1999; Menn et al., 2006; Xing et al., 2014). In the adult mouse, cortical OPCs exist in a regular, non-overlapping pattern that has been described as “tiled” in appearance (Hughes et al., 2013; Nishiyama et al., 2009). Further, OPCs can extend filopodia to continuously survey their environment, and they are capable of migrating to areas of demand to replace injured or dying cells (Hughes et al., 2013). Through a combination of growth and trophic factor exposure (Barres and Raff, 1993; Barres et al., 1994; Bansal et al., 1996) and epigenetic modifications (Marin-Husstege et al., 2002; Shen et al., 2005), OPCs can withdraw from the cell cycle, express mature OL proteins including myelin basic protein (MBP) and proteolipid protein (PLP), and differentiate into myelinating OLs. Mature OLs in the mammalian brain myelinate axons of diameters greater than 0.2 μm , but unmyelinated axons between diameters of 0.2 μm and 0.8 μm can also be found (Remahl and Hildebrand, 1982; Snaidero and Simons, 2014). Given an axon or axon-like substrate (e.g. polystyrene nanofibers or silica micropillars), oligodendroglial cells *in vitro* appear to follow an innate program of differentiation and myelin wrapping even in the absence of exogenous signaling (Lee et al., 2013; Mei et al., 2014). While this inherent propensity to wrap fibers of appropriate size is evident *in vitro*, the presence in the brain of appropriately sized yet unmyelinated fibers suggests a more complicated interplay of modulating factors *in vivo*.

2.2. Myelin and circuit function

Myelin sheaths channel electrical current down the axon instead of across the axonal membrane, thereby increasing signal conduction velocity (Hartline and Colman, 2007). By limiting ion flux across the axonal membrane to the unmyelinated nodes of Ranvier, fast, saltatory conduction of nerve impulses down the axon is possible (Tasaki, 1939). How do changes in myelin structure affect conduction velocity? Theoretical (Deutsch, 1969; Rushton, 1951) and computational (Smith and Koles, 1970) models predict that a g-ratio (the ratio of axonal diameter to the sum of the diameters of the axon and its myelin sheath) of 0.6–0.7 is optimal to maximize conduction velocity, given that internode distance remains constant, generating a roughly inverse parabolic relationship between g-ratio and conduction velocity. Very subtle changes in g-ratio within the steep portion of this parabolic curve can have dramatic effects on conduction velocity (Smith and Koles, 1970). As early as

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low
intensity

1905, this range of ideal g-ratio values was observed to correspond to the average g-ratio in spinal nerves (Donaldson and Hoke, 1905), a region that becomes fairly uniformly myelinated during development. Conduction velocity decreases as the g-ratio deviates in either direction from the predicted ideal value. However, when not all axons within a given functional circuit possess myelin sheaths of optimal thickness, small changes in myelin thickness to approach a more optimal g-ratio represent an opportunity to alter conduction velocity. This potential for “tuning” conduction velocity could thus alter circuit function. As discussed below, association projections in the human cerebrum are incompletely myelinated, highlighting the potential for such “tunability,” perhaps in an activity-dependent way.

How, then, would changing conduction speed affect neural circuit function, and ultimately cognition and behavior? Functioning of neural circuitry in behavior depends upon integration of excitatory and inhibitory inputs to a target neuron. In addition to the net effect of excitatory and inhibitory inputs, the timing of spike arrival also influences the probability of neuronal firing (Salinas and Sejnowski, 2000). As noted above, subtle changes in myelin microstructure can result in substantial changes in conduction velocity (Smith and Koles, 1970). Theoretically, conduction speed could prove critical to overall circuit function given that long-range communication in the CNS involves not only direct actions of excitatory projections on target cells, but also parallel competing pathways with countervailing influences involving recruitment of local inhibition. Even small changes in conduction velocity could thus become important for action patterns to be selected to exert appropriate influence upon output control regions. A more in-depth discussion of this complex topic can be found in recent, excellent reviews (Pajevic et al., 2014; de Hoz and Simons, 2015).

3. Myelination throughout life

3.1. Patterns of developmental myelination

As motor function increased in complexity throughout evolution, the need for faster motor responses to stimuli is thought to have driven the evolution of myelin, first appearing in cartilaginous fish (Zalc et al., 2008). Developmental myelination in altricial mammals occurs chiefly postnatally. In humans (a neurologically semi-altricial species), developmental myelination is particularly protracted and continues at least into the fourth decade of life (Yakovlev, 1967; Lebel et al., 2012; Miller et al., 2012). There is evidence that both the absolute level of myelination and the rate of myelination vary throughout life. The absolute volume of white matter (i.e. the gross regions of the brain mostly populated by myelinated axons) relative to total brain volume is highest in the fifth decade of life (Lebel et al., 2012), while the rate of new myelin formation is highest in the first decade (Benes et al., 1994). Myelination during development proceeds in predictable spatial and temporal patterns, as established by histological and neuroimaging studies (Brody et al., 1987; Yakovlev, 1967; Kinney et al., 1988; Lebel et al., 2012). Initial waves of myelin formation during early human postnatal development proceed in a proximal to distal fashion, from the central sulcus to the cerebral poles and from the cervical spinal cord caudally to the lumbar spinal cord; (this is why human infants can lift their heads prior to sitting up, which in turn occurs before walking (Gerber et al., 2010)). Thicker axons are myelinated before thinner ones (Hildebrand et al., 1993). In general, more complex brain regions corresponding to high-level cognitive and executive function are myelinated later than less complex regions devoted to basic neurological and homeostatic functions. As such, the brainstem is myelinated before the basal ganglia, which in turn becomes myelinated before cortical association fibers, required for

the highest level integrative and executive functions. Proximal regions myelinate before their distal counterparts, central before superficial, and sensory before motor. Areas that myelinate earliest become, on average, more heavily myelinated than areas that myelinate later (Glasser and Van Essen, 2011). Such patterns of myelination raise interesting questions about the possibility of region-specific differences in oligodendroglial lineage cell behavior.

3.2. Heterogeneity in myelination and myelin-forming cells

In addition to regional differences in the spatiotemporal patterns of myelination, basic parameters of adult myelin can vary considerably. Beyond variation in myelin sheath thickness (Baumann and Pham-Dinh, 2001), other properties that could have important effects on conduction velocity include internode length and premyelin axonal segment length (PMAS; the distance between axon hillock and the initial myelin internodal segment (Tomassy et al., 2014)). Deeper cortical layers are more densely myelinated than superficial layers, and mouse somatosensory cortical neurons in superficial layers have longer PMAS length than deeper layer neurons, correlating with a higher density of OLs (but not OPCs) in deeper layers than in superficial layers. At least in the mouse somatosensory cortex, superficial neurons exhibit intermittent myelination, with unmyelinated segments present between internodal segments that are much longer than the typical node of Ranvier. While the striking variation in myelination patterns along a single axon has been characterized only in mouse somatosensory cortex and visual cortex and its significance is not yet understood, it is possible that diverse myelin arrangements may represent another dimension by which myelination could influence neural function.

Subtle differences between oligodendrocytes have been appreciated for close to a century, as Pio del Rio Hortega, co-discoverer with Wilder Penfield of oligodendrocytes, classified them into four categories based on morphology and number of target axons (Del Rio Hortega, 1928). More recently, patterns of regional heterogeneity have begun to be elucidated. Oligodendrocytes in cerebral cortex make shorter internodes than their counterparts in the corpus callosum, subcortical gray matter, brainstem or spinal cord (Chong et al., 2012). There is striking internode length variability among oligodendrocytes in the same region of the CNS (i.e. cortex, corpus callosum, caudate putamen, cerebellum, brainstem, spinal cord) as well as among internodes belonging to one oligodendrocyte (Chong et al., 2012). Oligodendrocytes in the corpus callosum of rats also differ from cerebellar oligodendrocytes in their electrical properties: corpus callosum OLs have higher membrane resistance and conductance such that the modeled conduction velocity of an action potential in the corpus callosum is predicted to be half that of a similar axon in the cerebellum, presumably due to the need for rapid signaling in the cerebellum to facilitate cerebellar control of motor coordination (Bakiri et al., 2011).

Oligodendrocyte precursor cells also differ in their embryological origin (Kessaris et al., 2006). These cells are similar in their electrical properties, but dorsally derived cells preferentially myelinate dorsal tracts such as the dorsal corticospinal tract (which in contrast to humans runs in the dorsal spinal cord of the mouse) and rubrospinal tract (Tripathi et al., 2011). Elegant heterotopic transplantation experiments in the mouse reveal that the white matter niche allows both grey and white matter-derived OPCs to differentiate into mature OLs, while the grey matter niche promotes OL differentiation less. Furthermore, white matter-derived OPCs differentiate more frequently in both grey and white matter niches, suggesting that white matter OPCs exhibit cell-intrinsic differences from grey matter OPCs (Viganò et al., 2013). These findings are in

accordance with two earlier studies demonstrating that white matter allows for the constant generation of mature, myelinating OLs from OPCs, while the grey matter microenvironment appears to favor OPC differentiation less (Dimou et al., 2008), and that OPCs in white matter, but not in grey matter, are responsive to the mitogen platelet-derived growth factor (PDGF; Hill et al., 2013). Even within grey matter, while deeper cortical layers exhibit a greater density of oligodendrocytes than superficial layers, the density of OPCs in somatosensory cortex is constant throughout all layers (Tomassy et al., 2014). This suggests that the potential of OPCs to differentiate into myelinating OLs varies in a cortical layer-specific manner, either due to differing properties intrinsic to oligodendroglial lineage cells or to varying microenvironmental conditions (Tomassy et al., 2014). Given the striking heterogeneity in the behavior of oligodendroglial lineage cells, it is plausible that a similar diversity in the ability to adapt to changing neural activity patterns may exist in various regions of the nervous system.

4. Experience-dependent changes in myelin structure

4.1. Influences of experience during development

Myelination during childhood neurodevelopment may be sensitive to experience, as suggested by the lengthy period of developmental myelination in humans. Emerging evidence indicates that developmental myelination is particularly sensitive to experience during critical periods. Rhesus monkeys raised individually in a social isolation paradigm (i.e. housing separately from conspecifics) for 10 months exhibit a decrease in corpus callosum size, but not in other brain structures, which lasts at least six months after rehousing into a social setting. Further, these monkeys socially isolated during the juvenile period exhibit cognitive deficits in object learning tasks (Sanchez et al., 1998). Similarly, social isolation of developing mice for two weeks after weaning results in changes in prefrontal cortex oligodendrocyte morphology, gene expression and myelin sheath thickness. The consequences of isolation during this critical period do not recover after reintroduction of the affected mice into a more social environment (Makinodan et al., 2012). In contrast, social isolation at an age beyond this critical period does not produce lasting effects on prefrontal cortex myelin (Makinodan et al., 2012; Liu et al., 2012). Unilaterally clipping the whiskers of developing mice immediately after birth similarly results in decreased barrel cortex (the whisker somatosensory cortex) myelin density and myelin sheath thickness, but, unlike the critical period of juvenile social isolation on prefrontal cortex myelin, barrel cortex myelin sheath thickness recovers after whiskers are allowed to regrow (Barrera et al., 2013). Together, these data suggest that the effects of experience on myelin development may vary between brain regions.

Human neuroimaging studies support the idea that experience can affect myelin microstructure. Abused or neglected children may suffer cognitive deficits later in life, especially in attention, motivation, self-control, and emotional regulation (Egeland et al., 1983). While such sequelae of neglect are likely multifactorial in etiology, these deficits have been correlated with decreased fractional anisotropy (i.e. a measure of water diffusivity in diffusion-weighted imaging that correlates with myelination, fiber density, and axonal diameter; FA) in the limbic system white matter tracts central to some of these cognitive processes (Eluvathingal et al., 2006). Further, children aged 8–10 years who were poor readers exhibit hypomyelination of cerebral white matter (left anterior centrum semiovale) compared to good readers; however, after only 100 h of remediation in reading skills, FA in this region increased, suggesting a structural change in white matter accompanying increased utilization of this area through reading (Keller and Just, 2009). In piano

players, the extent of piano practice during childhood and adolescence was found to be positively correlated with FA of several regions of the pyramidal tract, including subcortical fibers in the frontal lobe and the corpus callosum. By adulthood, the extent of piano practice was only correlated with FA of the anterior limb of the internal capsule and the arcuate fasciculus. This restriction in anatomical regions exhibiting changes in FA correlated to piano practice with advancing age suggests that more distributed regions of white matter are sensitive to the experience of piano playing during childhood (Bengtsson et al., 2005). More work is needed to elucidate exactly which aspects of myelination are most sensitive to environmental deprivation or enrichment during development. Rodent models of experience-dependent myelin changes, while offering the benefit of mechanistic insight through genetic, pharmacological, and other manipulation, cannot fully recapitulate the complexity of life experience that may influence white matter in humans. Nonetheless, analogous decreases in myelin sheath thickness and white matter volume after similar patterns of experiential deprivation in animal and human studies suggest that at least some mechanisms of white matter response to experience are shared between the two species. These results showing lasting deleterious effects to myelination following early-life deprivation suggest that environmental stimulation during childhood neurodevelopment may be critical to establishing healthy myelin structure in regions of the brain important to emotional and cognitive function.

4.2. Influences of experience during adulthood

Even after development, mammalian CNS myelin appears to remain sensitive to experience-dependent structural change (as reviewed in Zatorre et al., 2012)). There is evidence that the acquisition of a new visuomotor skill, such as juggling, increases FA of white matter areas involved in visuomotor integration in adult humans (Scholz et al., 2009). Additionally, after two sessions of training in a balance task, the frontal subcortical white matter of adult human participants exhibited increased FA. FA continued to increase after a prolonged 6-week learning period, indicating a dose-responsive plasticity effect (Taubert et al., 2010). In animal models, training adult rats in a complex skilled reaching task resulted in increased FA in white matter subjacent to the contralateral sensorimotor cortex involved in the task, but not in the ipsilateral (untrained) hemisphere; both FA and myelin protein density as determined by immunohistochemical staining correlated with rate of learning the task (Sampaio-Baptista et al., 2013). Placing adult rats in an enriched environment, where rats were more cognitively stimulated than in standard cages, resulted in increased volume of the corpus callosum, total volume of myelinated fibers and their myelin sheaths within the corpus callosum (Zhao et al., 2012). These intriguing studies highlight how even brief, repetitive circuit activity can yield detectable changes in oligodendrogenesis and/or white matter structure (as in Gibson et al., 2014) and McKenzie et al. (2014), discussed below). Together, these data indicate that exposure to complex sensorimotor experiences may be sufficient in humans and rodents to increase myelination in relevant circuits, possibly to increase conduction velocity and alter circuit function.

Just as enriched environments are correlated with and can precipitate increased myelination in adults, deprivation may reduce myelin volume in adults. Social isolation of mice for the first time in adulthood results in thinning of prefrontal cortex myelin, decreased transcription of myelin-related genes and decreased social behavior. After rehousing the mice into a social environment, myelin protein-related gene expression and social behavior return to control levels (Liu et al., 2012). Deprivation of sensory

these monkeys

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information in developing rodent somatosensory cortex through whisker clipping decreases the survival of newly-divided OPCs by inducing apoptosis, resulting in a decrease in oligodendrocyte generation and overall oligodendrocyte density in the deprived cortex (Hill et al., 2014). While some reports show an increase in barrel cortex OPC proliferation following whisker clipping (Mangin et al., 2012), these cells may proliferate only transiently after the start of deprivation in response to OL loss (Hill et al., 2014). These data suggest that myelination is plastic and can respond to changes in the environment both during postnatal development and in adulthood, after developmental myelination has completed. Whether myelin plasticity in adults decreases during aging, as does synaptic plasticity (Burke and Barnes, 2006), is poorly understood.

4.3. Activity-induced changes within myelin-forming cells

How is experience transduced to oligodendroglial lineage cells? Experience can influence a multitude of physiological changes, including important endocrinological changes that could significantly affect myelin-forming cells. For example, social isolation-induced elevations in circulating glucocorticoids could in theory preferentially impair prefrontal cortex oligodendroglial lineage cells (Weiss et al., 2004; Stranahan et al., 2006; Grant et al., 2009; Alonso, 2000; Czéh et al., 2007; Banasr et al., 2007). Physical exercise such as running promotes oligodendrogenesis in the spinal cord (Krityakiarana et al., 2010). In addition, experience-dependent changes in white matter structure and oligodendroglial lineage cells may be related to changes in neuronal activity within an active circuit (Fig. 1). One of the first links between neuronal activity and changes in oligodendrocyte lineage cells was demonstrated in 1993, when it was shown that axotomy of the optic nerves or blockade of action potential propagation using the specific voltage-gated sodium channel blocker tetrodotoxin results in decreased proliferation of OPCs in the optic nerve (Barres and Raff, 1993). Similarly, silencing action potentials with tetrodotoxin results in

decreased myelination of axons in a peripheral nervous system *in vitro* coculture model (Schwann cells with dorsal root ganglion neurons) and in hypomyelination of the developing optic nerve in mice (Demerens et al., 1996). On the other hand, exposure to α -scorpion toxin, which increases neuronal firing, results in increased myelination in the coculture setting (Demerens et al., 1996), demonstrating the bidirectional plasticity of myelination in response to activity *in vitro*. As properties of myelin structure affect conduction velocity and circuit function, it is reasonable that different patterns of circuit activity could also have different effects on the adaptive response of myelin sheaths in that circuit. When Stevens et al. used electrically-induced neuronal activity to probe activity-dependent myelination *in vitro*, they discovered that low frequencies (0.1 Hz) of stimulation inhibit normal myelin formation while higher frequencies (1 Hz) do not (Stevens et al., 1998).

Elucidating the role of neuronal activity on oligodendroglial lineage cell dynamics *in vivo* has been more difficult until the recent development of new technologies for *in vivo* control of neuronal activity. While *in vivo* high frequency electrical stimulation (333 Hz) in the medullary pyramids of rats was found to result in a modest increase in proliferation and differentiation of dorsal corticospinal tract OPCs (Li et al., 2010), the potential neuro-inflammatory effects of deep electrode placement-related injury and the subsequent microenvironmental influence on oligodendroglial lineage cells complicates assessment of influences of neuronal activity on myelin-forming cells using this technique. The recent development of optogenetics, a technique in which light-sensitive ion channels can be selectively expressed in specific neuronal subpopulations and activated following exposure to specific wavelengths of light delivered at a distance from the target, allows for both spatial and temporal control of neuronal activity with minimal disruption of the neural microenvironment within standard experimental parameters (Arenkiel et al., 2007; Wang et al., 2007). Recently, *in vivo* optogenetic techniques were used in awake, behaving mice to investigate the role of neuronal activity on myelin-forming cells in the juvenile and young adult mouse CNS. Stimulation of 20-Hz neuronal activity in cortical layer V projection neurons of the premotor cortex (which typically fire at rates between 10 and 40 Hz) causes mice to exhibit complex motor behavior (ambulation in a unidirectional circle). This physiometric, behaviorally relevant neuronal activity results in activity-dependent OPC proliferation, differentiation into mature oligodendrocytes, and increase in myelin sheath thickness specifically within the active circuit. In contrast, pathological neuronal activity in the form of optogenetically-induced tonic-clonic seizures, does not induce OPC proliferation above levels found in identically-manipulated wild-type (no opsin present) controls (Gibson et al., 2014). While physiometric neuronal activity appears to play an important role in modulating the function of myelin and myelin-forming cells, at least in the premotor circuit, the relationship between newly generated oligodendroglial lineage cells and changes in myelin structure following activity is not yet understood. For example, while activity of neurons in the premotor circuit results in increases in both newly generated oligodendrocytes and myelin sheath thickness (Gibson et al., 2014), it is not yet clear whether newly-generated OLs contribute to myelin sheath remodeling or if preexisting OLs respond to increased axonal activity by adding additional myelin wraps in order to attain this increase in myelin thickness. Elegant OL-lineage fate-mapping work by Young and colleagues describing decreased internode length with age suggests that newly formed OLs do contribute to myelin remodeling over time, at least in the rodent brain (Young et al., 2013), but whether this is the primary mechanism by which activity-regulated myelin changes are achieved is presently unknown. Additionally, the molecular mechanisms governing activity-

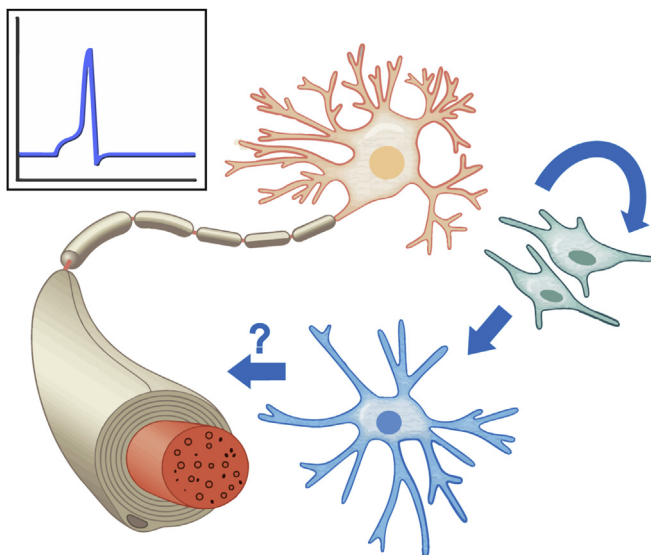


Fig. 1. Schematic of activity-dependent myelin plasticity. Neuronal activity (blue action potential trace) traveling down an axon (orange) can affect properties of myelin sheaths (grey) in the active circuit. Myelin is produced and maintained by cells of the oligodendroglial lineage, including oligodendrocyte precursor cells (teal) and oligodendrocytes (blue), which myelinate axons. OPC differentiation into oligodendrocytes can be influenced by neuronal activity (blue arrows), but the relative contribution of newly differentiated oligodendrocytes versus existing oligodendrocytes to the myelin sheaths of active axons is incompletely understood.

Adapted from illustration by Layla Lang.

dependent OPC proliferation, OL differentiation and activity-dependent effects on myelin microstructure in the postnatal CNS remain incompletely understood (see below).

Neuronal activity may also play a role in biasing which axons are myelinated during development. Studies modulating neuronal activity in developing (pre-myelinated) zebrafish support a role for neuronal activity in axon selection during myelin development, as blocking synaptic vesicle release with tetanus toxin reduces the number of myelinated axons and the number of myelin sheaths per oligodendrocyte in the zebrafish spinal cord (Mensch et al., 2015). Neurons silenced by tetanus toxin-expression or by Kir2.1 channel expression are specifically selected against for myelination (Hines et al., 2015). *In vivo* time lapse imaging in the developing zebrafish indicates that initiation of myelin wrapping (nascent sheath formation) is activity-independent, but that activity-dependent secretion of axonal factors biases active axons for myelination (Hines et al., 2015). Conversely, increasing overall neuronal activity pharmacologically with a GABA-A antagonist results in an increase in spinal cord myelinated axons and increase in myelin sheaths per oligodendrocyte (Mensch et al., 2015), although optogenetic upregulation of neuronal activity in the developing zebrafish spinal cord was not found to bias those axons for preferential myelination (Hines et al., 2015).

4.4. Heterogeneity and myelin plasticity

Potential differences in oligodendroglial lineage cells from different regions of the brain might underlie differences in the degree to which myelin and myelin-forming cells in different regions of the CNS respond to neuronal activity. The previously unappreciated diversity in arrangement of internodal segments recently discovered by Tomassy and colleagues and discussed above may have functional consequences not only on signal conduction velocity and circuit function, but also on the capacity of myelin adaptation to neuronal activity (Tomassy et al., 2014). Additionally, heterogeneity in either oligodendroglial cells or neurons could explain the observation of myelin plasticity in some

regions but not in others.

Association projections appear particularly prone to exhibit plastic changes in myelin. Rats trained on a skilled reaching task exhibited higher FA, correlating with increased myelination, only in association fibers subjacent to the sensorimotor cortex corresponding to the trained limb (Sampaio-Baptista et al., 2013). The mouse premotor circuit has an association limb projecting from the ipsilateral primary motor cortex across the corpus callosum to the contralateral motor cortex, as well as a corticofugal limb projecting down the corticospinal tract (Fig. 2). Optogenetically-induced neuronal activity in the mouse premotor circuit recruits oligodendroglial lineage cell proliferation only in the association fiber limb, but not in the corticofugal limb (Fig. 2; Gibson et al., 2014). Whether this preferential association fiber pattern is observed in other circuits and in other species remains to be seen. Association fibers, which are among the last brain regions to myelinate (Brody et al., 1987) and end up with a lesser degree of myelination (Glasser and Van Essen, 2011), may be particularly amenable to myelin plasticity. On the other hand, corticofugal projection fiber myelin of the motor circuit may not share the same sensitivity to neuronal activity as do cerebral association fibers, at least in the rodent premotor circuit at the ages examined. While the reasons for this are unclear, it is possible that altering conduction velocity by remodeling myelin may not be functionally advantageous in all parts of the CNS.

5. Mechanisms regulating activity-dependent myelin changes

5.1. Cellular mechanisms

Elegant work suggests a link between motor skill learning and myelination by newly-generated OLs: McKenzie and colleagues showed that mastering a new motor skill (e.g. running on a complex mouse wheel with irregularly spaced rungs) results in a transient spike of OPC proliferation followed by significantly increased density of newly-generated OLs. Further, when OPCs were genetically manipulated to prevent differentiation into OLs

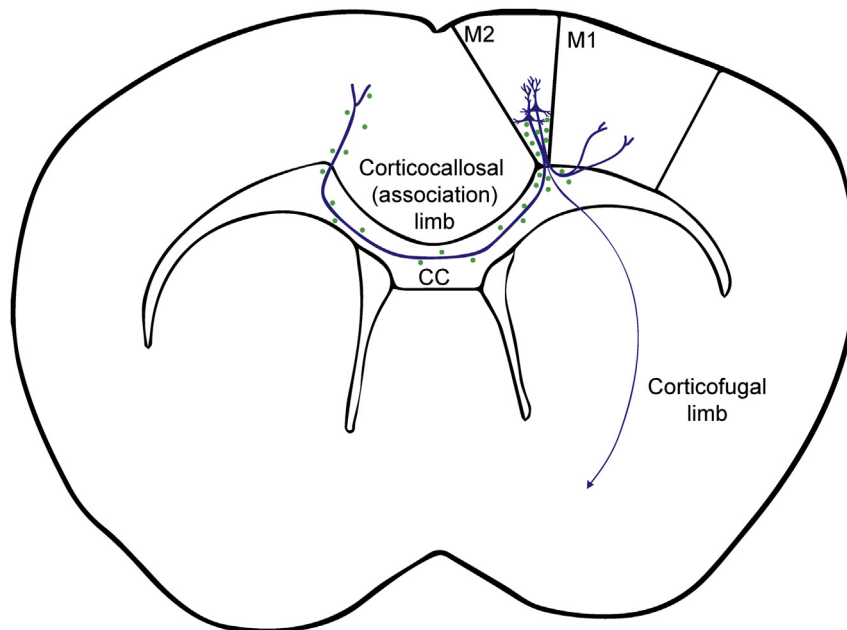


Fig. 2. Heterogeneity in oligodendroglial lineage cell response to neuronal activity. Optogenetic stimulation at a physiologic frequency of layer V pyramidal neurons in premotor cortex (M2) of Thy1::ChR2 mice induces proliferation of oligodendroglial lineage cells (green dots) in the association fiber, or corticocallosal, limb of the premotor circuit (blue neurons projecting across the corpus callosum [CC]), but not in the corticofugal limb.

during adulthood, mice were unable to master running on the complex wheel, suggesting that learning a new motor skill requires the generation of OLs in mice (McKenzie et al., 2014). Additionally, in the nearly fully-myelinated adult rodent optic nerve, OPCs continue to actively proliferate and generate new OLs, suggesting that newly-born oligodendrocytes may be actively engaged in remodeling the myelin sheaths of the optic nerve by either replacing dying oligodendrocytes or intercalating new internodes into existing myelin infrastructure (Young et al., 2013). This constant proliferative capacity of OPCs may represent the cellular basis by which plasticity can be quickly initiated after only a brief stimulus (as in Gibson et al. (2014)). While OPCs seem to constantly turn over in adult organisms, it appears that their sensitivity to exogenous factors in the environment may be highest during a three to eight-day window after cell division. During this time period, demyelinating injury and damage to oligodendrocytes in the mouse brain causes more rapid OPC differentiation, while barrel cortex hypoactivity resulting from whisker clipping causes a decrease in the new differentiation of oligodendrocytes and survival of newly-divided OPCs (Hill et al., 2014). It should also be noted that once formed, new oligodendrocytes exhibit a limited time period in which they can establish myelin sheaths, at least in the developing zebrafish (Czopka et al., 2013).

Non-human primates also exhibit evidence of new OL generation throughout life despite developing adult levels of myelinated fiber density by the age of sexual maturity, suggesting that newer cells may be involved in adaptive responses (Miller et al., 2012; Peters, 2009). Analysis of the incorporation of ^{14}C (a byproduct of atomic weapons testing in the 20th century) in the corpus callosum of humans reveals that ^{14}C levels in myelin mirror concentrations contemporary with the time of death, indicating that human myelin undergoes robust turnover throughout life. On the other hand, nuclei in the corpus callosum were found to contain ^{14}C that more closely matches ^{14}C levels during the individual's childhood, which may suggest less cellular turnover in the human corpus callosum in adulthood compared to rodent or non-human primates, and highlights the potential importance of existing mature oligodendrocytes in ongoing human corpus callosum myelin remodeling (Yeung et al., 2014). Future work will need to clarify the extent to which human myelin remodeling proceeds from newly-generated oligodendrocytes versus existing oligodendrocytes.

5.2. Molecular mechanisms

Several hypotheses regarding the molecular mechanisms underlying myelin plasticity have been investigated in recent decades. Initial studies in the hypoactive rat optic nerve postulated the release of a soluble factor (such as PDGF or glutamate) by electrically active neurons either directly or indirectly through astrocytes, causing oligodendroglial proliferation (Barres and Raff, 1993). Demerens and colleagues hypothesized that the release of a soluble factor from electrically active axons or the expression of a promyelogenic surface factor on active neurons led to increased myelination (Demerens et al., 1996). The latter idea is supported by evidence linking hypomyelination resulting from stimulation of low-frequency neuronal activity and decreased expression of the cell adhesion molecule L1 (Stevens et al., 1998). Functional adenosine receptors have been identified on OPCs *in vitro*, and pharmacological blockade of these receptors results in decreased Ca^{2+} signaling within OPCs (Stevens et al., 2002). On the other hand, adding adenosine to OPC cultures decreases their proliferation, promotes differentiation into oligodendrocytes, and induces a myelinating phenotype. This suggests that purinergic signaling

might play a role in transducing axonal action potentials to OPCs and promoting lineage progression (Stevens et al., 2002).

Another mechanism implicated in facilitating the transition from oligodendroglial progenitor to immature oligodendrocyte is PI(3,4,5)P3 signaling to the AKT/mTOR pathway (Tyler et al., 2009). AKT/mTOR activity results in increased OL membrane wrapping and myelination without changes in OL population size (Goebbels et al., 2010; Flores et al., 2008). An innovative recent study of plasma membrane dynamics during OL-to-axon contact and myelin wrapping confirmed that PI(3,4,5)P3 is necessary for the growth of the inner tongue of the ensheathing oligodendrocyte membrane. Further, the presence of an electric field around cultured oligodendrocytes was found to be sufficient to promote the membrane trafficking processes required for myelination (Snaidero et al., 2014). This elegant work raises the possibility that the electrical consequences of neuronal activity, such as accumulation of cations released from axons propagating action potentials, could trigger processes of myelination and may thus contribute to activity-dependent myelin plasticity.

Intriguing work has identified oligodendroglial lineage cells that express AMPA (Bergles et al., 2000) and NMDA (Káradóttir et al., 2005, 2008) receptors and form glutamatergic synapses with neurons (Mangin et al., 2008; De Biase et al., 2010). This body of evidence raises the possibility that direct neuron-glia signaling through synaptic connections could underlie myelin plasticity, potentially via the release of glutamate from axonal vesicles and the subsequent synthesis of MBP (Wake et al., 2011). However, conditional deletion of an NMDA receptor subunit in mouse oligodendroglial lineage cells did not cause deficits in developmental myelination *in vivo*; of note, possible compensation through AMPA receptor upregulation was observed (De Biase et al., 2011). More recent work has attempted to reconcile the diverse hypotheses surrounding glutamatergic signaling in oligodendroglial lineage cells. Using a neuron-OPC coculture system, Lundgaard et al. propose two “modes” of myelination: an activity-independent mode that occurs in the absence of glutamate, and an activity-dependent mode that occurs when axonal glutamate release triggers NMDA receptor currents in oligodendrocytes and results in accelerated and increased myelination. The key to switching between modes appears to be the presence of neuregulin (NRG) and BDNF, which facilitate an increase in oligodendroglial NMDA receptor currents and leads to increased myelination (Lundgaard et al., 2013). Signaling through BDNF and its receptor TrkB appears to be crucial for OL differentiation and myelination or re-myelination (Wong et al., 2013; VonDrán et al., 2011; Fulmer et al., 2014; Tshiperson et al., 2015), making BDNF – known to be secreted in an activity-dependent manner (Zafra et al., 1990; Tao et al., 2002; Matsuda et al., 2009) – a compelling candidate for an activity-induced promyelogenic signal. One potential mechanism for this may be through the metabotropic glutamate receptor activation-induced secretion of astrocytic BDNF, as has been shown to promote re-myelination following cuprizone-induced demyelination (Fulmer et al., 2014). Astrocytes, a source of BDNF along with neurons (Zhang et al., 2014; Rudge et al., 1995; Dougherty et al., 2000; Korte et al., 1995) have already been implicated in activity-dependent myelination in the peripheral nervous system (PNS) via ATP and LIF (Ishibashi et al., 2006). Since both BDNF and NRG via TrkB and ErbB signaling, respectively, can alter NMDA receptor expression in oligodendrocytes, these two pathways may compensate for each other if one is disrupted (Lundgaard et al., 2013). Such redundancy in the pathways mediating NMDA-receptor mediated myelination highlights the potential importance of activity-related myelination.

Neuregulins were first implicated in modulating myelination within the PNS in relation to the ability of Schwann cells to detect

axonal diameter. Neuregulin1 (NRG1) proteins are divided into three protein families: type I, II, and III. NRG1 type III was shown to play an essential role in determining ensheathment of PNS axons as well as myelin sheath thickness and internodal length in mice (Taveggia et al., 2005; Michailov et al., 2004; Chen et al., 2006). Paracrine secretion of NRG1 appears to affect myelination in a concentration-dependent manner with low concentrations being promyelinating and high concentrations inhibiting myelination (Syed et al., 2010). Determining the role of NRG and ErbB signaling in myelination of the CNS is complicated by the fact that all three groups of NRG1 receptors are expressed in the CNS, whereas only ErbB2 and ErbB3 are expressed in the PNS, and both *nrg1* and *erbb* null mutants are embryonic lethal. Some transgenic models of altered ErbB signaling in oligodendrocytes exhibit hypomyelination and a decrease in conduction velocity of CNS axons associated with changes to oligodendrocyte differentiation and morphology (Kim et al., 2003; Roy et al., 2007). Underscoring the heterogeneity discussed above, while mice haploinsufficient for NRG1 type III exhibit hypomyelination of the brain, myelination in the optic nerve and spinal cord of these mice is normal. The observed hypomyelination in NRG1^{+/-} mice was independent from changes in oligodendrocyte differentiation, suggesting potential mechanistic differences underlying NRG/ErbB mediated myelination (Taveggia et al., 2008). While other reports suggest that developmental myelination of the CNS might be independent of NRG/ErbB signaling, upregulation of this pathway appears to stimulate hypermyelination of cortical axons and increased density of myelinated fibers relative to oligodendrocyte density (Brinkmann et al., 2008), indicating that in the CNS NRG/ErbB signaling may play a distinct role in adaptive myelination. In accordance with these previous studies and the potential role of NRG in NMDA-mediated myelination, Makinodan and colleagues found that social isolation of mice during development (P21–35) results in hypomyelination of the prefrontal cortex, reduced prefrontal cortex-mediated cognitive function, and reduced NRG expression. This myelin and cognitive phenotype could be recapitulated through loss of ErbB3 signaling during the same developmental time period (Makinodan et al., 2012), suggesting a role for NRG signaling in experience-dependent regulation of myelination. NRGs may play varying roles in the development and maintenance of PNS and CNS myelination, as well as in developmental and adaptive myelination (Brinkmann et al., 2008), and could be an important molecular switch in determining the type of myelination occurring within a circuit based on temporal and spatial determinants. Thus, the dual-mode hypothesis proposed by Lundgaard and colleagues (Lundgaard et al., 2013) implicating NRG as a potential molecular switch to NMDA-mediated myelination is intriguing in that it might underlie the difference between the basal level of myelination that happens irrespective of neuronal activity during development and myelination that is adaptive and tunable.

5.3. Epigenetic mechanisms

Responses by oligodendroglial lineage cells to neuronal activity may be due in part to altered epigenetic regulation of gene expression patterns and chromatin state within these cells following neuronal activity or hypoactivity. Repressive histone H3 trimethylation at lysine 9 (H3K9me3) marks are increased during OPC differentiation to the OL stage. Ablating H3K9 histone methyltransferase activity restricts differentiation and enhances the electrical excitability of the immature OL membrane, suggesting that epigenetic mechanisms may be important in regulating oligodendroglial response to neuronal activity (Liu et al., 2015). Following social isolation, mouse prefrontal cortex OLs in areas of hypomyelination show decreased heterochromatin levels,

increased histone acetylation marks, and decreased H3K9me3 marks, suggesting a deprivation-induced development of a chromatin state that impairs myelination (Liu et al., 2012; Shen et al., 2008). Consistently, following optogenetic stimulation of neuronal activity in the mouse premotor circuit, proliferating cells exhibit an activity-dependent increase in H3K9me3 and a concurrent decrease in H3 acetylation, suggesting the initiation of a differentiation program (Gibson et al., 2014). Epigenetic mechanisms regulating oligodendroglial lineage progression, such as histone deacetylase activity and its crosstalk with the Wnt pathway (Marin-Husstege et al., 2002; Cunliffe and Casaccia-Bonneli, 2006; Ye et al., 2009), Brg1-dependent chromatin remodeling (Yu et al., 2013), and micro RNA (miRNA) and long intervening non-coding RNA (lncRNA) regulation of OL maturation (Mills et al., 2015; Dugas et al., 2011), are beginning to be elucidated. Connecting these pathways to the molecular consequences of neuronal activity will shed further light on the molecular basis of adaptive myelination.

While several seemingly independent mechanisms have been proposed to explain activity-dependent changes in myelin-forming cells, it is important to note that no one mechanism detailed above can explain these diverse and complicated phenomena. Adaptive changes in myelin most likely result from a combinatorial effect of several molecular mechanisms identified as important to myelin remodeling. It is also possible that specific mechanisms predominantly contribute to distinct processes of activity-dependent myelination (e.g. OPC proliferation, OL differentiation, myelin wrapping). Despite the progress made in the past few decades to discern the role of neuronal activity in myelination, a great deal remains unknown regarding the precise mechanisms mediating activity-regulated oligodendroglial lineage cell proliferation, differentiation and changes in myelin microstructure.

6. Outlook

Mounting evidence supports the concept that myelin is a dimension along which experience directs structural and functional adaptive change, at least in some regions of the nervous system. As the molecular mechanisms mediating plastic changes in OPC proliferation, differentiation and myelin microstructure become increasingly clear, so too will the roles such changes may play in both healthy and pathological neurological function. Neuronal activity-dependent changes in myelin-forming cells (Gibson et al., 2014) may prove critical in cognitive (McKenzie et al., 2014) and psychiatric (Makinodan et al., 2012; Liu et al., 2012) function. The potential circuit level implications of even small changes in myelin microstructure and subsequent conduction velocity imply that mechanisms of myelin plasticity must be tightly regulated. Given the robust mitogenic effect of neuronal activity on oligodendroglial lineage precursor cells, dysregulation or “hijacking” of activity-dependent proliferation signals could play a role in the initiation and/or growth of some forms of brain cancer, particularly those that arise from or share molecular characteristics with oligodendroglial lineage cells (Venkatesh et al., 2015). On the other hand, a deeper understanding of the mechanisms governing activity-dependent myelin changes may be harnessed for regenerative strategies in diseases of myelin.

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